



BIOTECH REVOLUTION IN A BOTTLE: THE SCIENCE BEHIND YOUR SKINCARE

I never expected to learn about biotechnology from my bathroom shelf. I never imagined I'd question what retinol does or why niacinamide is needed for my skin. Yet here I am, realizing that skincare is not just about beauty, but also deeply connected to modern biotechnology.

The global skincare market is projected to reach over \$108 billion by 2032 (Yingngam, 2024). It's a field where biochemistry, biotechnology, genetics, and dermatology intertwine. Skincare is no longer just about fixing skin problems; it's about science working with your skin. These ingredients, which might have been developed in a biotech lab, target the skin's anatomy and interact with the skin's biological function to improve skin health.

According to Gomes et al. (2020), many active compounds like retinoids, hyaluronic acid, vitamin B3, and peptides are developed through biotechnological and chemical processes, helping strengthen the skin barrier, reduce inflammation, and boost collagen. Many of these ingredients are derived from natural sources, where microorganisms are utilized for their bioactive compounds, and fermentation techniques yield beneficial extracts. For instance, niacinamide (Vitamin B3) can be synthesized through microbial fermentation. Hyaluronic acid is often produced by streptococcus fermentation (Desai & Mane, 2023). One of biotechnology's greatest innovations, recombinant DNA technology, is also widely used in this sector. A piece of DNA coding for a skin-enhancing protein, like collagen boosters, can be inserted into a microorganism and then grown in labs. Peptides, common in anti-aging serums, are introduced to skincare via this method.



In 2025, skincare has become a global trend, a form of preventive medicine to maintain and improve skin health and to prevent acne over time. Many brands, such as The Ordinary and CeraVe, have gained popularity worldwide for promoting clinically proven active ingredients. Brands are investing in biotech innovation or developing their research labs, opening new career paths for future biotechnologists to invent smarter and effective ingredients for our skin health. However, this progress also raises questions like– is this high-quality, bio-engineered skincare truly accessible and affordable to everyone, or is it just a luxury? What about the environmental impact of mass-produced skincare? How sustainable is the sourcing of these high-value ingredients? Are we overloading our skin with too many actives? Could we be just disrupting the skin's natural microbiome with overuse of chemical actives? Or are we simply following trends without understanding the science behind them?

At this point, a single drop of these serums now carries the story of years of scientific dedication and experimentation by the researchers. Here, different fields of science meet self-care, presenting a world where biotechnology is not just used to create medicines and to cure illness, but also to enhance everyday life.